BigQuery SQL Practice

Exploring the Google Analytics Sample Dataset

About the project

This document contains my solutions to a set of analytical tasks written in Google BigQuery (Standard SQL, run in the BigQuery sandbox). The queries cover session counting, filtering by date with table wildcards, grouping and ranking, working with nested and repeated fields (`RECORD` / `UNNEST`), and metric interpretation based on the official schema documentation.

About the dataset

The queries use the public dataset `bigquery-public-data.google_analytics_sample`. It contains obfuscated Google Analytics 360 data from the Google Merchandise Store, an online e-commerce store selling Google-branded products. The data covers the period from **1 August 2016 to 1 August 2017** and is stored in one table per day, named `ga_sessions_YYYYMMDD`.

Grain of the data: one row = one session. Session-level attributes are stored as regular or nested columns, and everything that happens inside a session (pageviews, events, transactions) is stored in the repeated `hits` field.

Column reference

Full description of every column in the export table: Google Analytics Help, *[UA] BigQuery Export schema [Legacy]*
https://support.google.com/analytics/answer/3437719?hl=en

Field	Type	Description
`fullVisitorId`	STRING	Unique visitor identifier
`visitId`	INTEGER	Session identifier (unique only per visitor)
`date`	STRING	Session date in YYYYMMDD format
`totals`	RECORD	Session-level totals: visits, hits, pageviews, timeOnSite, transactions, totalTransactionRevenue
`trafficSource`	RECORD	Traffic source: source, medium, campaign, keyword
`device`	RECORD	Device data: deviceCategory, browser, operatingSystem
`geoNetwork`	RECORD	Location data: country, region, city
`hits`	RECORD, REPEATED	Individual interactions in the session (pageviews, events, transactions)

Note: all queries read the sharded tables with the wildcard `ga_sessions_*`. The wildcard combines all daily tables into one, and the `_TABLE_SUFFIX` pseudo-column can be used to limit the date range.

Task 1. Count the total number of sessions

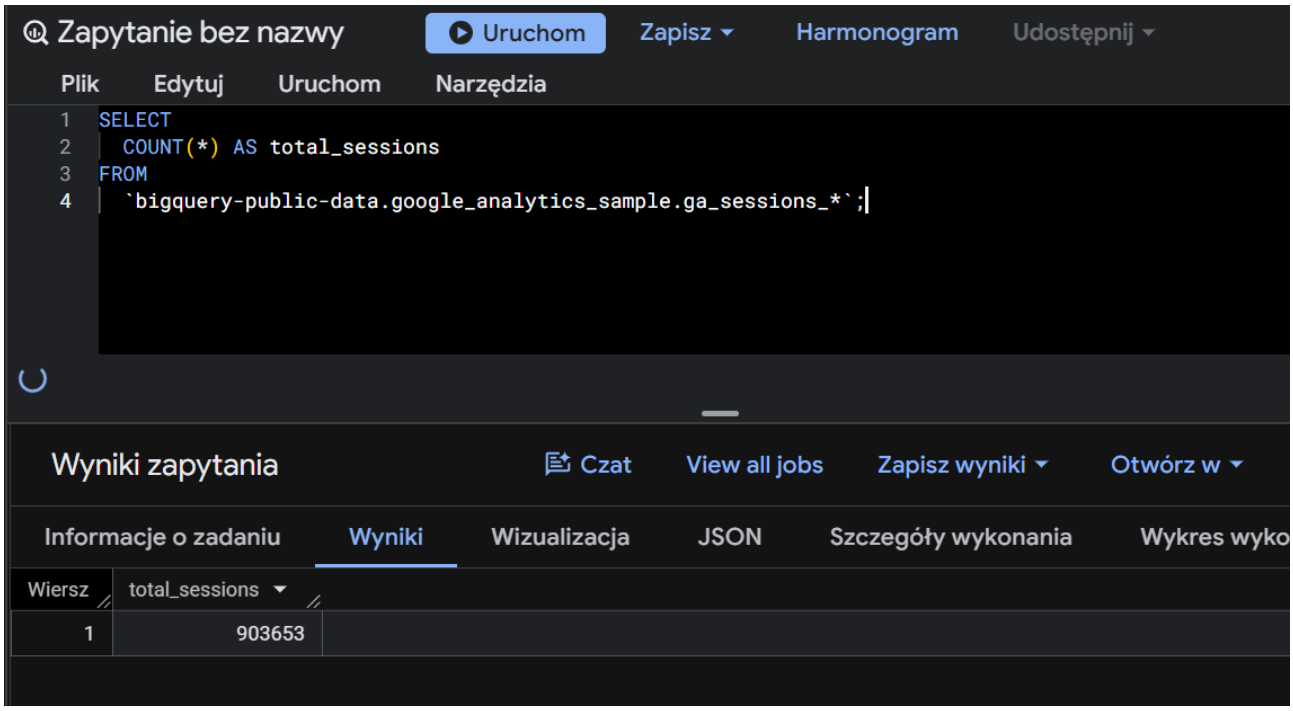
Approach

Each row in a `ga_sessions_*` table represents one session, while the pageviews and events of that session are nested inside the repeated `hits` field. Counting rows therefore counts sessions. The wildcard `*` reads all daily tables at once.

SQL query

```
SELECT
  COUNT(*) AS total_sessions
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`;
```

Result



Task 2. Count sessions on a specific date (31 July 2017)

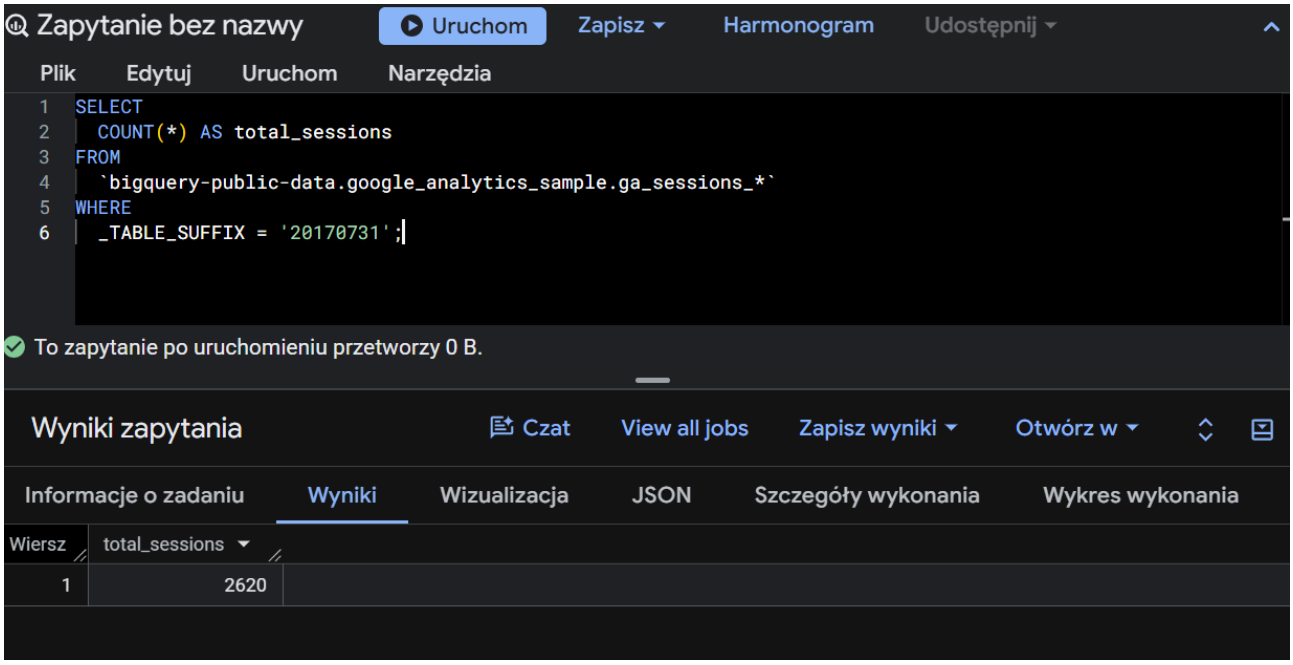
Approach

Filtering on the `_TABLE_SUFFIX` pseudo-column makes BigQuery scan only the table for that day instead of the whole year, which reduces processed data and cost. The suffix is a string in `YYYYMMDD` format.

SQL query

```
SELECT
  COUNT(*) AS total_sessions
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`
WHERE
  _TABLE_SUFFIX = '20170731';
```

Result



Task 3. Sessions by device category

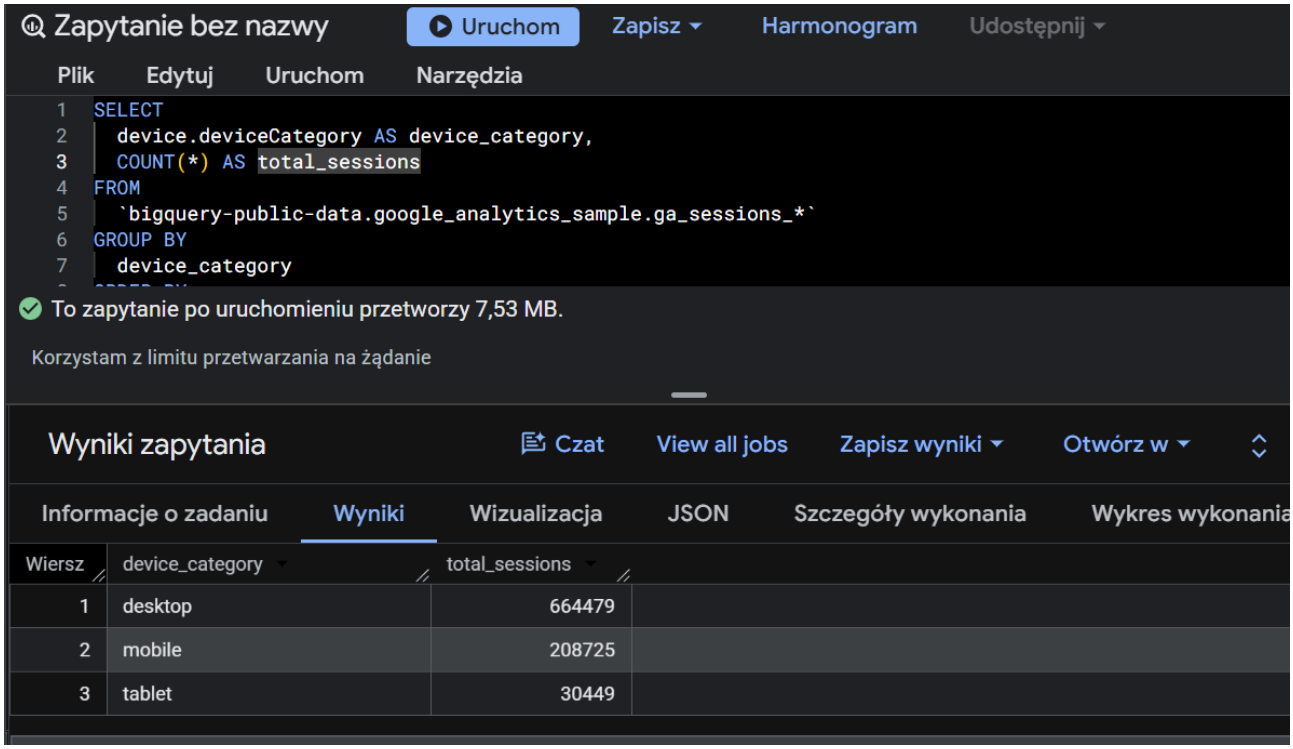
Approach

`device` is a nested RECORD (not an array), so its fields are accessed with dot notation and no `UNNEST` is needed. Possible values: desktop, mobile, tablet.

SQL query

```
SELECT
  device.deviceCategory AS device_category,
  COUNT(*) AS total_sessions
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`
GROUP BY
  device_category
ORDER BY
  total_sessions DESC;
```

Result



Task 4. Top 5 traffic sources by number of sessions

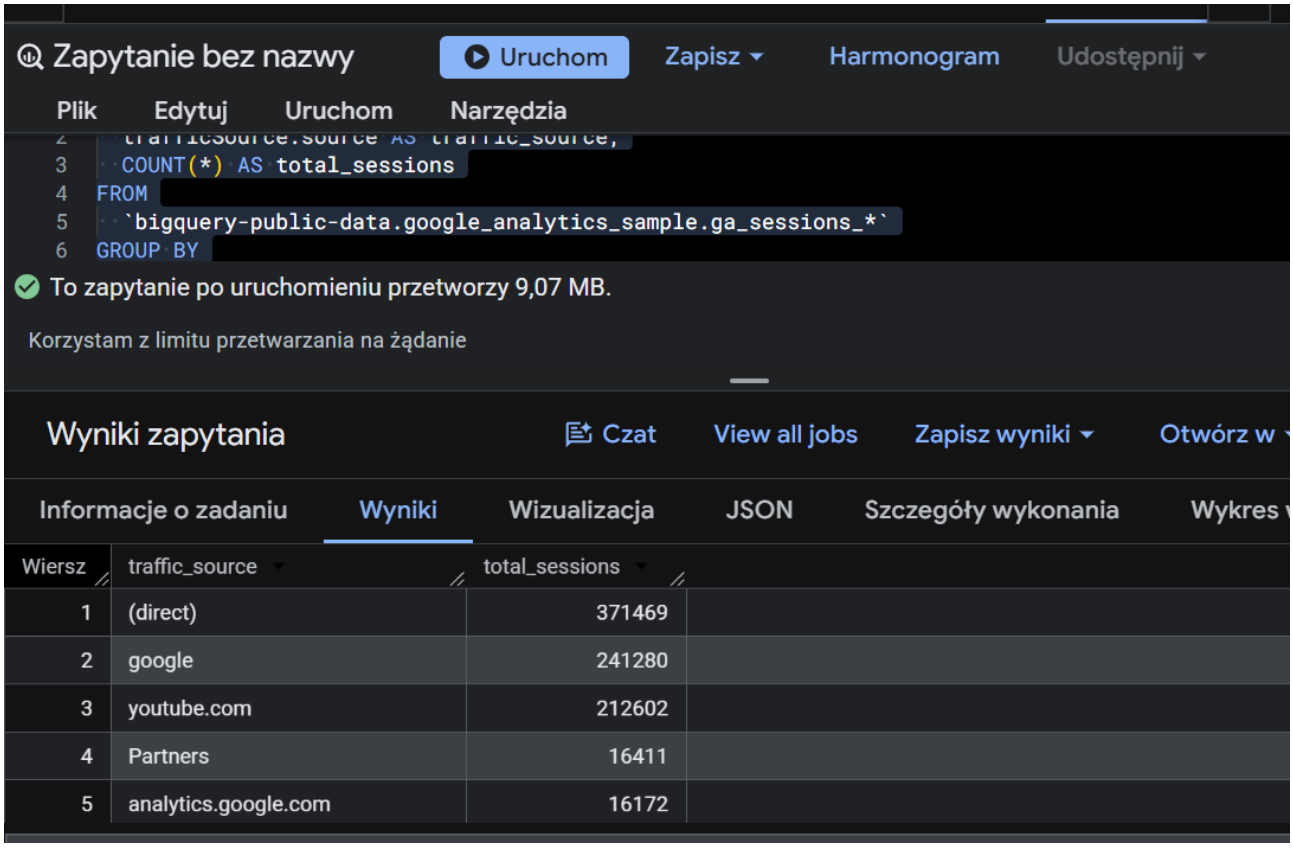
Approach

Sessions are grouped by `trafficSource.source`, sorted in descending order and limited to five rows. The value `(direct)` represents visits with no referring source.

SQL query

```
SELECT
  trafficSource.source AS traffic_source,
  COUNT(*) AS total_sessions
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`
GROUP BY
  traffic_source
ORDER BY
  total_sessions DESC
LIMIT 5;
```

Result



Task 5. Top 10 pages by number of pageviews

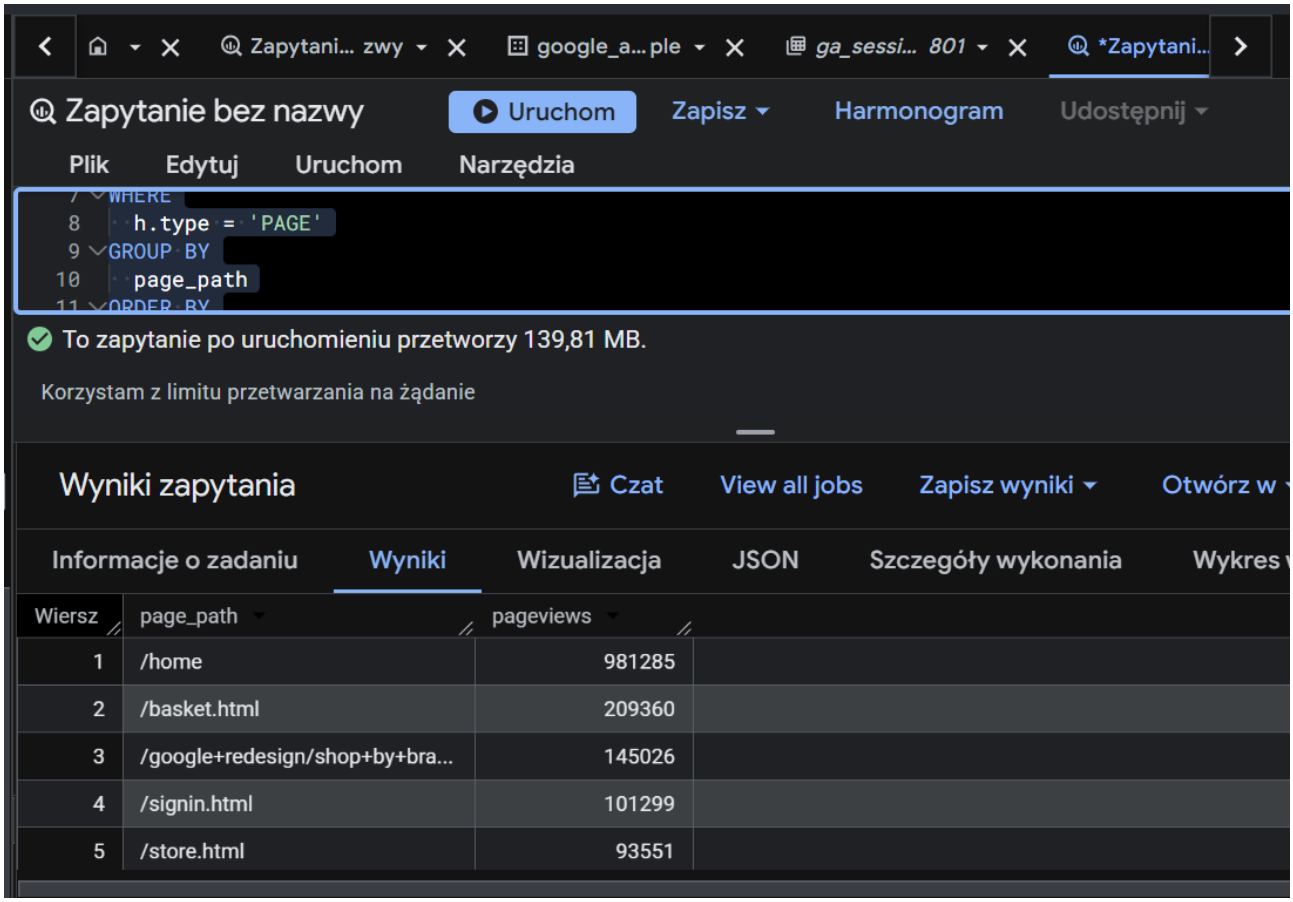
Approach

`hits` is a repeated RECORD, so it is flattened with `UNNEST`: after that, each row is a single hit and `COUNT(*)` counts hits instead of sessions. The filter `type = 'PAGE'` keeps only pageviews (excluding events, transactions, etc.).

SQL query

```
SELECT
  h.page.pagePath AS page_path,
  COUNT(*) AS pageviews
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`,
  UNNEST(hits) AS h
WHERE
  h.type = 'PAGE'
GROUP BY
  page_path
ORDER BY
  pageviews DESC
LIMIT 10;
```

Result



Task 6. Average session duration (seconds)

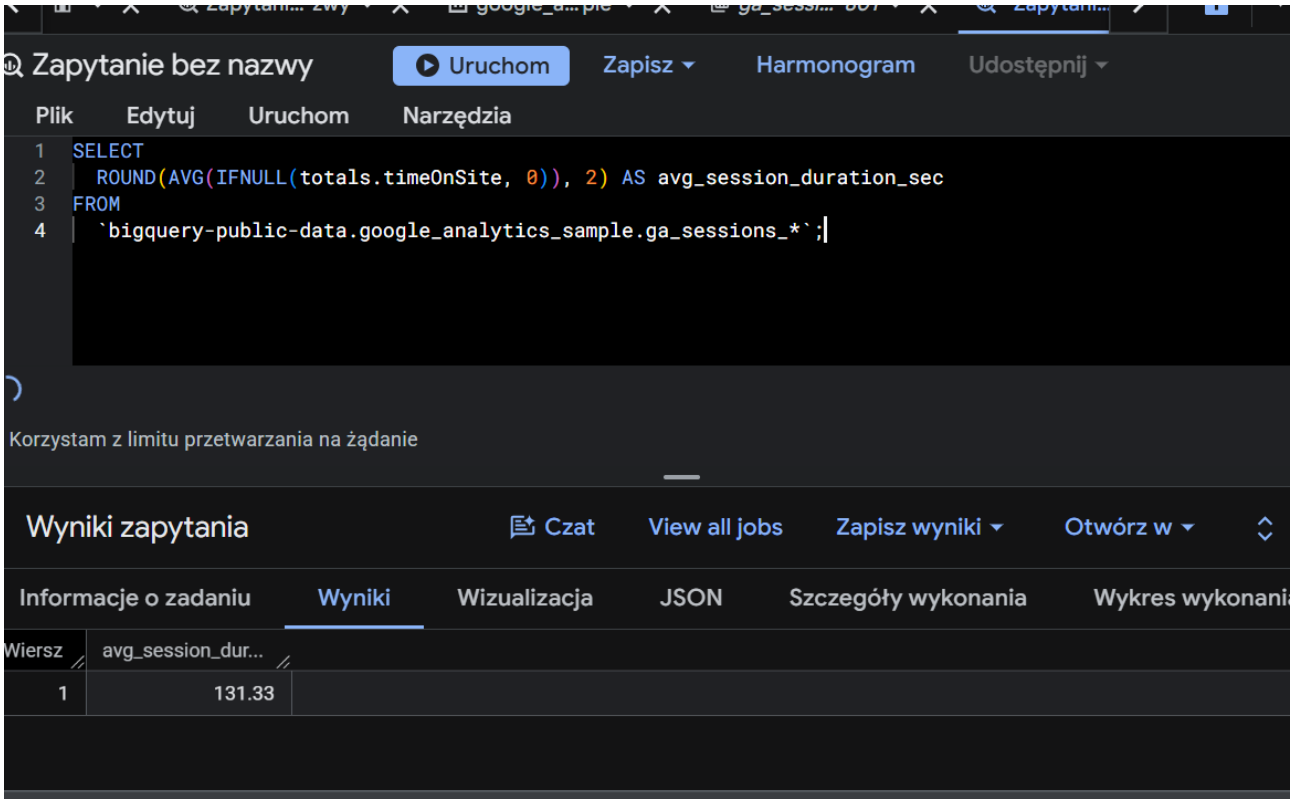
Approach

`totals.timeOnSite` stores session duration in seconds. It is `NULL` for single-hit sessions (bounces), because Google Analytics cannot measure time on site without a second interaction. `IFNULL(..., 0)` counts such sessions as 0 seconds, so the average is calculated across **all** sessions rather than only those with a measurable duration.

SQL query

```
SELECT
  ROUND(AVG(IFNULL(totals.timeOnSite, 0)), 2) AS avg_session_duration_sec
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`;
```

Result



Task 7. Top 5 countries by number of sessions

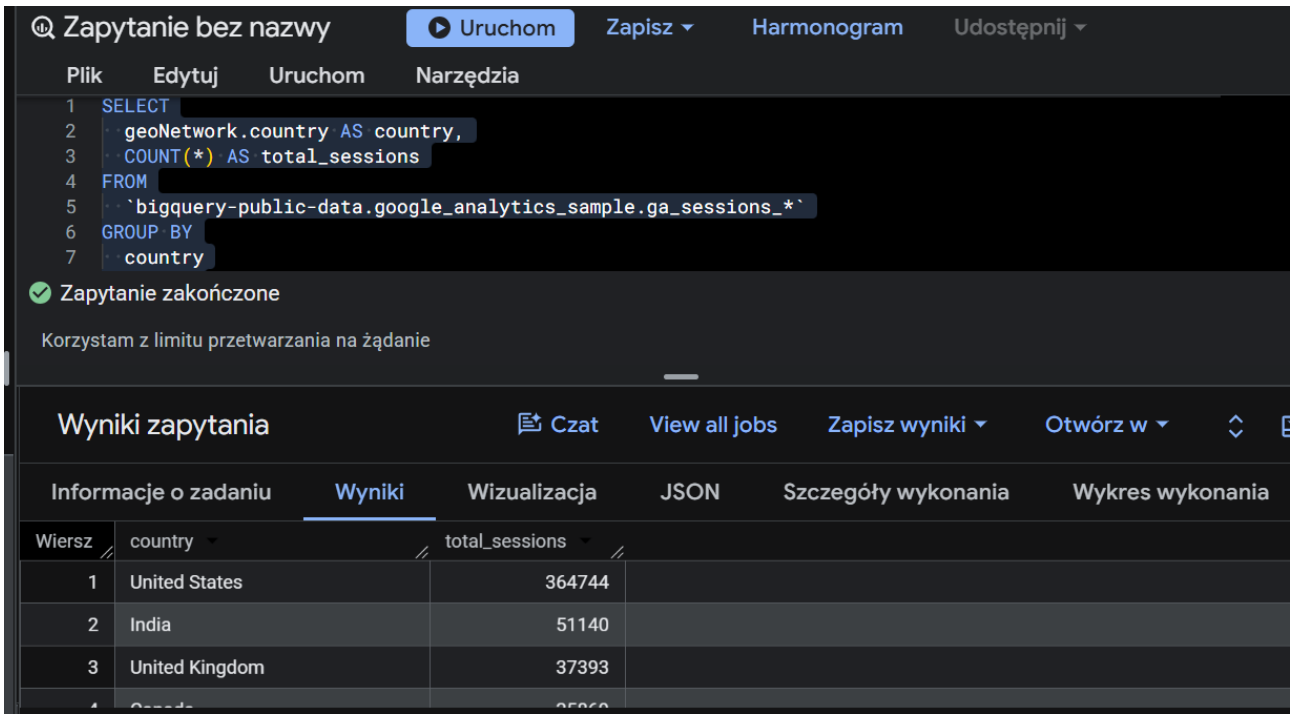
Approach

`geoNetwork` is a nested RECORD, so the country is read with dot notation. Sessions whose location could not be determined are labelled `(not set)`.

SQL query

```
SELECT
  geoNetwork.country AS country,
  COUNT(*) AS total_sessions
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`
GROUP BY
  country
ORDER BY
  total_sessions DESC
LIMIT 5;
```

Result



Task 8. Total transactions and total revenue

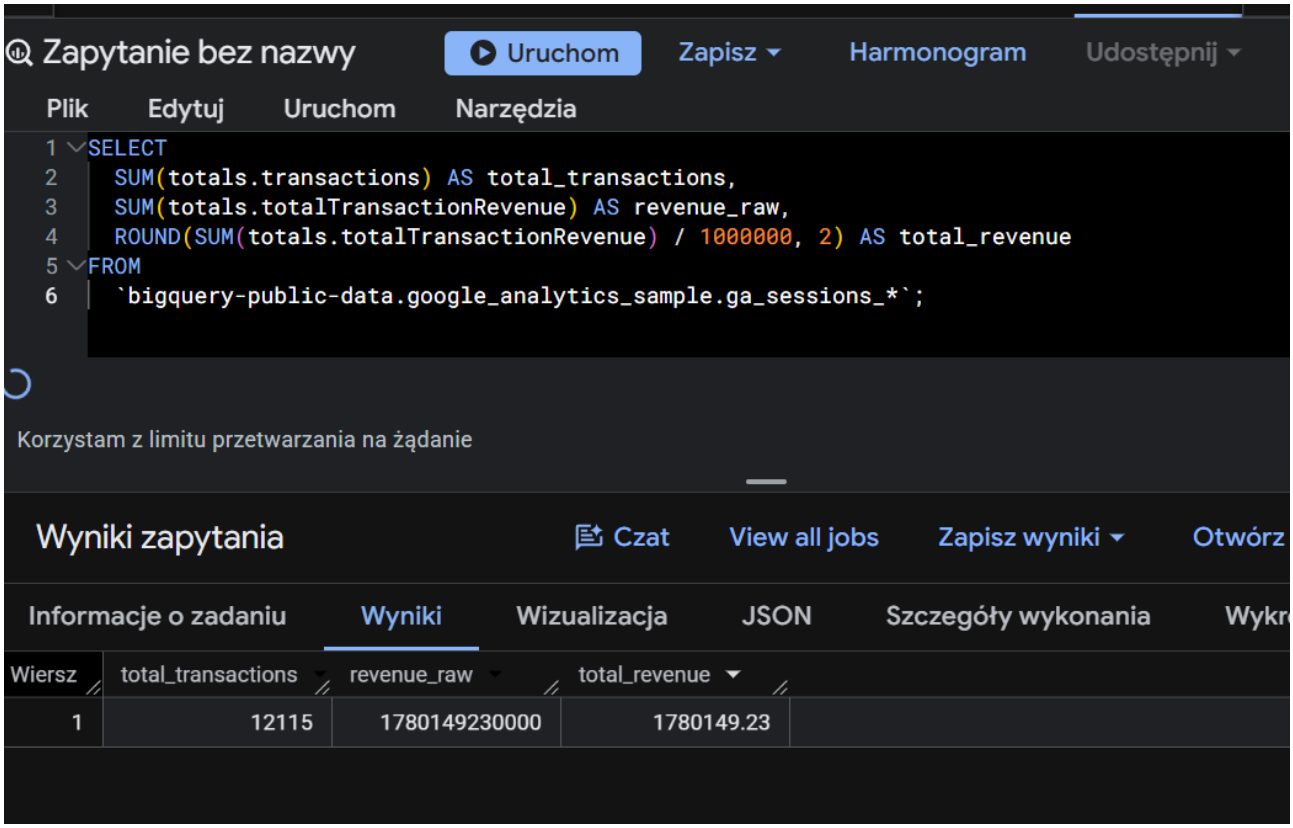
Approach

According to the schema documentation, `totals.transactions` is the number of ecommerce transactions in the session, and `totals.totalTransactionRevenue` is the total transaction revenue **multiplied by 10^6**. To get values in the same units as shown in the Google Analytics interface (USD), the sum is divided by 1,000,000. `SUM` ignores NULLs, so sessions without purchases do not affect the result.

SQL query

```
SELECT
  SUM(totals.transactions) AS total_transactions,
  SUM(totals.totalTransactionRevenue) AS revenue_raw,
  ROUND(SUM(totals.totalTransactionRevenue) / 1000000, 2) AS total_revenue
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`;
```

Result



Source

Column descriptions: https://support.google.com/analytics/answer/3437719?hl=en